KIRMAN SHAHAB

Senior Full Stack Engineer | TypeScript · Node.js/NestJS · Next.js · Distributed Systems · AI Agents

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SUMMARY

Senior full stack engineer who builds the hard parts of SaaS: multi-tenant isolation, job queues that don't melt, billing and metering, and AI agents that safely touch real customer data. ≈4,300 commits over 3 years leading a 111-module multi-tenant CRM (NestJS, Next.js, MongoDB, Redis/BullMQ); previously blockchain (Casper) and React Native.

EXPERIENCE

Technical Architect, Full Stack Development — Jul 2023 – Present

XisLabs, Lahore, Pakistan

- **Technical architect** for **Maple Connect**, a **multi-tenant SaaS** CRM for lending and mortgage teams: **111** NestJS modules, ~4,000 TypeScript files, ~1,340 test specs. Top contributor with **≈4,300 commits** across 9 production apps.
- Also **business analyst** and **project manager** for a small team: requirements, specs, sprints, delivery.
- Architected **multi-tenant data isolation** with request-scoped tenant context and guards across all 111 modules, plus **RBAC** (**JWT**, **OAuth 2.0**) and plan-based entitlements enforced on API and client.
- Built **event-driven job processing** on **BullMQ** and **Redis** for campaigns, drip sequences, cron automations and bulk imports. Designed a **global Redis rate budget** shared by every worker after an email-provider suspension; **zero suspensions** afterwards.
- Removed a **cron thundering herd** (every past-due campaign firing at 07:00:05) with jittered scheduling, and moved scheduled email off the provider onto BullMQ delayed jobs with pause/resume.
- Shipped a **Model Context Protocol (MCP)** server exposing CRM operations to AI agents, with tool calling across **3 LLM providers** (Claude, OpenAI, Gemini). Built per-use **token metering** and a wallet, and fixed an uncharged-calls bug with a data migration.
- Delivered **real-time** features on **Socket.IO** with a Redis adapter for **horizontal scaling**, and integrated 7 platforms: **Stripe** (subscriptions, webhooks), **Twilio** Voice/SMS, Retell AI, SendGrid, **AWS S3/SES**, Google and Outlook calendars.
- Ran **CI/CD** on **GitHub Actions** and **Docker** with **Playwright** e2e tests; added a TypeScript type-check gate to an SWC build that was silently shipping 17 type errors, and fixed a Next.js build peaking at 7.64 GB on an 8 GB builder.
- Built **MedusaJS** headless storefronts (Meilisearch, multi-region, 2 payment gateways) and a **React Native** app for WoodUImagine.

Blockchain & Full Stack Developer — Jan 2022 – Jul 2023

Scytale Labs, Remote

- Built JS/TS client libraries for the **Casper Network** blockchain (casper-js-sdk): deploys, signing, state queries.
- Designed and tested **GraphQL mutations** that connected smart contracts to the backend and frontend, end to end.
- Built a **React Native** app (iOS/Android) with wallet connectivity and transaction signing.
- Built **MERN** stack services and dashboards (MongoDB, Express, React, Node.js).

Freelance Frontend Developer — 2021

Self-employed, Remote

- Built responsive, mobile-first **React** apps for e-commerce and small business clients, from requirements to delivery.

SELECTED PROJECTS

Maple Connect, multi-tenant lending CRM: leads, pipelines, campaigns, AI agents, billing. *· NestJS, MongoDB, BullMQ, Next.js*

Raven Gray, headless storefront: faceted instant search, multi-region, dual payment checkout. *· MedusaJS, Next.js, Meilisearch*

AI Voice Agent Platform, voice-agent SaaS: self-serve signup, Stripe checkout, automated Twilio numbers. *· Next.js, Retell AI*

TECHNICAL SKILLS

Backend	TypeScript, Node.js, NestJS, Express, REST, GraphQL (Apollo/Yoga), Socket.IO, BullMQ, Mongoose
Frontend & Mobile	React, Next.js (App Router), React Native, React Query, Zustand, Redux Toolkit, Tailwind CSS
Data & DevOps	MongoDB, Redis, Meilisearch, Docker, GitHub Actions, AWS (S3, SES), DigitalOcean
AI & Blockchain	MCP, tool calling, Claude, OpenAI, Gemini, Vercel AI SDK, Casper Network, Rust (basic)
Architecture	Multi-tenant SaaS, RBAC, event-driven design, job queues, rate limiting, webhooks, horizontal scaling
Leadership	Technical architecture, business analysis, requirements gathering, project management, Agile/Scrum

EDUCATION & LANGUAGES

BSc in Computer Science, Virtual University of Pakistan, Lahore — 2016 – 2020

Junior Full Stack Development Program, Gaper, Lahore — 2021

Languages: English (professional), Urdu (native)